

# A bounded, observable, auditable 7-agent DAG.

Six concentric layers. Seven parent agents. Twenty-two sub-agents. Built on the same Bedrock + Claude Sonnet 4.6 + MCP stack Cognizant standardised on for the Anthropic partnership announced November 4, 2024. Every layer has a single responsibility. Every agent has a single competency. Every LLM call is graded, retried, and audited. **The architecture is the compliance posture** — not bolted on, not an afterthought.

7

PARENT AGENTS  
LANGGRAPH DAG

22

SUB-AGENTS  
BOUNDED RESPONSIBILITY

6

ARCHITECTURE LAYERS  
DATA → OPS

98

LLM CALLS / CASE  
VERIFIED LIVE-RUN

\$1.01

REAL PER-CASE COST  
VS \$1,500 MANUAL

§ 4 Six concentric layers · single responsibility per layer

READ TOP-DOWN · TOP-DOWN DEPENDENCY ONLY

LAYER 6 · OPS PLANE  
runbooks · checklists · evidence packs

Bedrock migration runbookDemo-day checklistCMS-0057-F scorecardROI calculatorAnticipated Q&ASMOKE\_TEST\_RESULTS.md

LAYER 5 · SURFACE PLANE  
React+Vite+TS · standalone HTML demo

React 18 + Vite 5 + TypeScript strictTailwind + lucide-reactSSE client · live agent traceStandalone HTML showcase · 13 routes  
Tweaks panel · live demo control

LAYER 4 · GATEWAY PLANE  
LLM provider abstraction · cost-aware routing

LLMClient interfaceBedrockClient · prodAnthropicClient · fallbackOpenRouterClient · devCost router · haiku ↔ sonnet  
Token-budget guardrailPer-tenant cost ledger

LAYER 3 · RUNTIME PLANE  
LangGraph orchestration · SSE bus · reflection

LangGraph DAG · statefulFastAPI + uvicorn · asyncSSE streaming · /cases/:id/streamWorker · case\_runner  
Reflection loop · self-grade + retryStructured-error escalation

LAYER 2 · AGENT PLANE  
7 parents · 22 sub-agents · pure functions

7 parent agents22 sub-agentsPydantic v2 input/output schemas.txt prompts (no inline)  
SchemaGuardrail · TokenBudgetGuardrail · PHIGuardrailPer-agent contract testFramework Grader · 5-field GraderScore

LAYER 1 · DATA PLANE  
PostgreSQL · row-level security · hash-chained audit

casesagent\_runsllem\_invocationsdecisionsappealsaudit\_ledger · SHA-256 chainphireductionsRLS per tenant · 10-yr retention

§ 5 7-agent LangGraph DAG · 22 sub-agents · bounded competencies

FLOW: TOP-LEFT → BOTTOM-RIGHT · MODEL PER AGENT ANNOTATED

1 Clinical Extractor  
FHIR R4 → TYPED SNAPSHOT  
— biomarker\_specialist  
— phi\_sanitizer  
— fhir\_resource\_validator  
Sonnet 4.6 · Haiku 4.5

2 Policy Retriever  
HYBRID KEYWORD + SEMANTIC + LLM  
— keyword\_filter  
— q\_business\_retriever  
— llm\_reranker  
— citation\_resolver  
Q Business + Sonnet 4.6

3 Necessity Reasoner  
PER-CRITERION MET/NOT\_MET/AMBIGUOUS  
— criterion\_splitter  
— evidence\_matcher · : parallel  
— confidence\_calibrator  
Sonnet 4.6 (parallel) · Haiku 4.5

4 Decision Composer  
VERDICT + RATIONALE + CITATION CHAIN  
— verdict\_synthesizer · det.  
— rationale\_writer  
— citation\_linker  
Sonnet 4.6 · Haiku 4.5

5 Denial Forecaster  
PAYER-SIDE DENIAL PROBABILITY + REASONS  
— probability\_estimator  
— reason\_predictor  
— appeal\_path\_recommender  
Sonnet 4.6 · Haiku 4.5 × 2

6 Appeals Drafter  
NCCN-CITED LETTER · STRUCTURED ARGUMENTS  
— counter\_evidence\_finder  
— nccn\_reference\_specialist  
— letter\_composer  
Sonnet 4.6 (long-JSON 8k) · Haiku 4.5

7 Patient Communicator  
6TH-GRADE · EMPATHIC · ZERO PHI  
— action\_step\_writer  
— empathy\_layer  
— reading\_level\_tuner · det.  
Sonnet 4.6 (patient-voice) · Haiku 4.5

OUTPUT BOUNDARY  
TriZetto submission  
structured envelope · BAA-covered transit · CMS-0057-F § IV.D · payer adjudication API

ExtractionRetrievalReasoningDecisionForecastAppeals (DENY only)CommunicatorBoundary : Parallel fan-out · det. = deterministic, no LLM

§ 6 · COGNIZANT ALIGNMENT MATRIX · WHY THIS FITS

ANTHROPIC STACK · NOV 4, 2024

NEURO-SAN AAOSA

HEALTH SCIENCES VERTICAL

AGENT FOUNDRY

TRIZETTO INTEGRATION

KIRO IDE

Bedrock + Claude Sonnet 4.6 + MCP — exactly the stack Cognizant standardised on

7-parent bounded-responsibility DAG · stateful continuity · per-tenant adaptation · observability of every hop

Oncology PA wedge · named 2026 focus area · 38% YoY oncology PA volume growth

Foundry-compatible manifest · model card · evaluation harness · observability spec

Provider→payer structured-envelope submission · production-ready hand-off · mock + real adapters

Spec-driven development · .kiro/specs/ on disk · proves Cognizant IDE story

AWS REFERENCE · AP-SOUTH-1 · MUMBAI

Bedrock-native, region-pinned to Pune.

BedrockQ BusinessECS FargateRDS PostgresKMSCloudWatch  
S3SESALBVPC

Provider abstraction: LLM\_PROVIDER=bedrock|anthropic|openrouter env-var flips client. Region pinned: ap-south-1 (Mumbai) — 11ms latency to Pune onsite. Cost-aware: Haiku 4.5 for grading + lightweight extraction · Sonnet 4.6 for reasoning. BAA-ready: healthcare workloads in HIPAA-eligible services only.

✓ INDEPENDENTLY VERIFIED · 2026-05-05 LIVE-RUN AUDIT  
authrex.xzashr.com  
scan QR → standalone interactive demo · 13 routes · 3 demo paths (APPROVE / REFER / DENY) · zero backend

AUTHREX  
TEAM AEROFYTA  
COGNIZANT  
TECHNOVERSE  
2026